

Assignment-03

Submission Guidelines:

- add your Student-ID, Assignment Number and Date of Submission at the very beginning
- follow the sequence given below
- add required screenshots containing the command you run and output you produced
- Preserve your data playground, ie: database, tables or data records after the completion of the assignment as a proof of work, I might want to see your work in your PC or wherever you have committed your work

Submission Deadline: April 13, 2026

Kick-off: Create a new database named *lab_5*.

Data Framing: Create a 3 new tables to store existing students information, students' score who registered for DBMS and OOP course using the given DDL as followings:

- ***students(roll_number, name, email, gender)***

```
CREATE TABLE students (roll_number integer AUTO_INCREMENT  
PRIMARY KEY, name text, email text, gender text);
```

- ***dbms_marksheet(roll_number, grade)***

```
CREATE TABLE dbms_marksheet(roll_number integer PRIMARY KEY,  
grade text);
```

- ***oop_marksheet(roll_number, grade)***

```
CREATE TABLE oop_marksheet(roll_number integer PRIMARY KEY,  
grade text);
```

Data Entry:

- insert sample data records of 50 students in ***students*** table

```
INSERT INTO students (name, email, gender)  
SELECT  
    CONCAT('student', @row := @row + 1) AS name,  
    CONCAT('student', @row, '@uiu.ac.bd') AS email,  
    IF(RAND() < 0.5, 'Male', 'Female') AS gender  
FROM (  
    SELECT @row := 0
```

```
FROM information_schema.COLUMNS a
CROSS JOIN information_schema.COLUMNS b
LIMIT 50
) numbers
ORDER BY RAND();
```

- insert sample grades of students who enrolled DBMS:

```
INSERT INTO dbms_marksheet (roll_number, grade) values (1,
"A+"), (2, "A"), (16, "A"), (20, "D"), (45, "C");
```

- insert sample grades of students who enrolled OOP:

```
INSERT INTO oop_marksheet (roll_number, grade) values (11,
"B-"), (12, "A"), (16, "A+"), (20, "B+"), (50, "A");
```

Data Retrieval:

1. Show the students' roll number along with their grades of DBMS and OOP for those students who enrolled both in DBMS and OOP courses this semester.
2. Show the students' roll number along with their grades of DBMS and OOP for those students who enrolled at least in DBMS.
3. Show the students' roll number along with their grades of DBMS and OOP for those students who enrolled at least in OOP.
4. Show the students' roll number along with their grades of DBMS and OOP for those students who enrolled in any of the courses.

Bonus Task (Advanced):

5. Show the name, email and roll number of all of the students along with their DBMS and OOP grades for those students who enrolled in any of the courses.
6. Show the students' roll number along with their grades of DBMS and OOP for those students who enrolled only in DBMS.
7. Show the students' roll number along with their grades of DBMS and OOP for those students who enrolled only in OOP.